

Accelerating Brain Stroke Prediction: Utilizing Parallel Computing and Clinical Features

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Brain stroke is a significant health issue, affecting 16 million people annually. It occurs when blood flowing to the brain is stopped or when a blood vessel bursts, leading to difficulties in movement and speech, and potentially resulting in death. Thus, early recognition of stroke symptoms is crucial for reducing severity, highlighting the importance of accurate prediction. This study investigates the impact of parallel computing techniques on hyperparameter tuning for stroke prediction using clinical features. With the increasing prevalence of strokes globally, accurate prediction models are essential for effective prevention and strategies. This research applied parallel computing to optimize hyperparameters. Also the results reveal a significant enhancement in predictive accuracy and computational efficiency compared to traditional methods, also underscores the potential of integrating advanced computing techniques in medical research, paving the way for more reliable and timely stroke prediction. Moreover, this study also outline future research directions to further improve stroke prediction models, emphasizing the necessity of larger and more diverse datasets.

Additional Key Words and Phrases: Parallel Computing, HPC, Brain Stroke, Prediction

ACM Reference Format:

SADIA RUHAMA, Brac University, Dhaka, Bangladesh . 2024. Accelerating Brain Stroke Prediction: Utilizing Parallel Computing and Clinical Features. 1, 1 (September 2024), 7 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Background

Brain Stroke is a serious concern, affecting 16 million people each year and about 5.5 million of those people die, it mainly happens when blood flowing in the brain is suddenly stopped or when a blood vessels in the brain bursts which makes it difficult to talk or move, and even results in death [17] [5]. According to the World Health Organization (WHO), one of the leading cause of death and disabilities around the world is due to stroke, therefore, recognizing the early signs of stroke can minimize the severity of brain stroke, [13] as a result, the need for accurate prediction is an essential. Furthermore, there are two main types of stroke: Ischemic Stroke and Hemorrhagic Stroke, Ischemic occurs when blood flowing in the brain suddenly stops due to blockage of blood vessels, and Hemorrhagic stroke happens when a blood vessels bursts which leads to bleeding in or around the brain, also strokes are often accelerated with unhealthy lifestyles and health conditions, such as, high blood pressure, diabetes, heart diseases, and many others [10].

On that account, early detection is crucial as it allows for quick treatment while reducing the long term damage by decreasing the severity of the stroke so many recent studies objective is focusing how accuracy can be increased, along with interpretability of the model [19]. There are many Machine Learning model that can be implemented to predict brain stroke, such as Random Forest (RF), Decision Tree (DT), and Naive Bayes (NB), however, a recent study showcases

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that Random Forest model is the most accurate one, with 95 percent accuracy [2]. Another study also utilized three machine learning algorithms: Random Forest (RF), Support Vector Machine (SVM), and Decision Tree (DT) and they also claimed that Random Forest has the highest accuracy with 95 percent [1]. Aside from that, several studies also utilized different methods to predict the outcome based on clinical images by using a deep learning method called convolutional neural network (CNNs) and demonstrated promising results [9]. However, current imaging techniques such as MRI and CT are effective but costly, time-consuming, and sometimes even harmful so recent studies utilizes microwave imaging combined with machine learning to transform stroke diagnosis, as a result, offering a viable alternative to traditional imaging techniques [11]

In brain stroke prediction, parallel computing is crucial as it helps maintain large volumes of datasets effectively and in less time, as machine learning models need to process a lot of data so by using multiple processors to divide the work, parallel computing speeds up tasks like training models and finds the best hyperparameters which makes predictions faster and more accurate [5]. Parallel computing is a method which allows multiple computations to be carried out simultaneously, as a result, allowing large volumes of data to be carried out efficiently [14]. Recent studies also utilizes massive parallel computing and advance computer modeling to create detailed images of the brain [20]. Furthermore, as brain stroke is one of the leading cause of death, with more than 15 million newly affected every year, the need for effective treatment and accurate prediction facility is essential, however, traditional methods such as MRI, CT scan are not suitable for frequent bedside monitoring due to their size, cost, and radiation exposure, hence, a study utilized a programmable system-on-chip (PSoC) that integrates an ARM-based CPU with a Field-Programmable Gate Array (FPGA) to offload heavy computations while achieving a fivefold speed increase in processing without compromising accuracy [16]. Following this, imbalanced dataset are also a major problem during model building for brain stroke prediction, hence, in order to improve the predictive outcome from imbalanced dataset, PMFE-SN is highly effective even when the number of samples is very low, which traditional methods cannot outsmart, also PMFE-SN outperforms state-of-the-art techniques in various evaluation metrics as well [12].

Brain stroke has a significant negative impact on the society, hence utilizing the recent technology to achieve the best possible outcome is the goal of many studies these days, a recent studies showcases that age, heart disease, average glucose level, and hypertension are the key indicators for detecting stroke in patients [4]. Furthermore, clinical features are vital because they provide a comprehensive view of the patient's condition, clinical features such as, age, sex, medical history, neurological assessment scores, brain imaging data are essential because these features combined with radiomics data has been used to create an accurate predictive model [7]. Moreover, in recent times, current methods for predicting brain strokes focuses on using imaging technologies, such as CT and MRI where the goal is to quickly detect whether a stroke is ischemic or hemorrhagic, and identify the damaged areas of the brain, however, there are certain limitations such as time sensitivity as stroke detection and treatment needs to happen very quickly it can be challenging with current methods [8].

Therefore, this study aims to utilize parallel computing techniques to enhance the predictive accuracy, along with computational efficiency for brain stroke prediction by using clinical features. By implementing advanced machine learning algorithms such as the Random Forest classifier while leveraging a parallel computing framework, this paper seeks to address the challenges posed by large datasets and improve the speed and reliability of stroke predictions.

2 Methodology

The purpose of this study is to speed up the prediction of brain stroke using Parallel Computing and HPC techniques and the approaches that the paper followed are described as follows which includes data preprocessing, architecture of the model, and lastly evaluating the model.

2.1 Data Acquisition and Preprocessing

The dataset used in this study was extracted from Kaggle [18], a popular online platform that has numerous collections of healthcare datasets. Initially, the paper started with 5000 patient samples where each sample represents various clinical features, for instance, age, hypertension, BMI, glucose level and many others which are crucial for predicting the disease. Following this, the paper handled the missing values using parallel computing. The BMI and smoking status features contain missing values which are essential for the prediction, hence to overcome the limitation, the paper enabled parallel processing across all available processors, as a result, enhancing the efficiency of the computational tasks. Also the missing values in the BMI feature was replaced by mean value of the BMI column which was accomplished by using the Joblib Library, and for the smoking status feature, the missing values were replaced by "Unknown" which helped keeping the data intact without removing potential patients records from the dataset. Furthermore, the dataset contains categorical variables that were handled using one-hot encoding, allowing the data to be transformed into numerical values and help Machine Learning model to interpret the results efficiently.

2.2 Model Architecture and Tuning

Feature selection is one of the crucial steps during model construction and the study's dataset mostly contains all relevant features, therefore, the model utilized all the features, excluding the target and patient ID. Furthermore, the paper utilized a machine learning model, Random Forest Classifier, as it is highly effective in handling different types of features. Also recent studies state that Random forest can be effectively used for stroke prediction [6]. Afterwards, in order to improve the efficiency of the model, the paper used GridSearchCV for hyperparameter tuning, as hyperparameter tuning involves adjusting the parameter of the model to optimize its performance, hence, to achieve the best possible outcome the paper accounted for this approach. The GridSerch is employed to systematically explore the different combinations of hyperparameter, and to increase the computational efficiency the GridSearch was utilized along with parallel computing to speed up the hyperparameter tuning process which uses all available cores of the computer rather than evaluating each parameter combination sequentially. It distributes the work across multiple processors, as a result, increasing efficiency and helps making better decisions regarding which tasks to assign to which resources [15]. Lastly, the model used 80 percent of the samples for training and 20 percent for testing.

2.3 Evaluating the Model

In order to evaluate the performance of the Random Forest Classifier model, the study used different metrics, for instance, accuracy, precision, recall, and F1-score were calculated to assess the accuracy of the model. The metrics also provided a comprehensive view of the model's ability to correctly predict brain stroke cases. For the hyperparameter tuning, GridSearchCV was employed which uses all the available processing cores and explores different combinations of hyperparameters. As a result, reducing the time required to find the optimal set of hyperparameters, and finally improving the overall performance of the model. In conclusion, the model achieved an overall accuracy of 93.93 percent on the test set, which demonstrates its ability to predict brain stroke with a high level of precision.

3 Findings

The paper showcases that we can quickly and efficiently predict brain strokes solely based on clinical features with the help of parallel computing techniques. Furthermore, the paper witnessed a high predictive accuracy when using Random Forest Classifier by achieving an impressive accuracy of about 93.93 percent on the test dataset [1], proving that machine learning is effective in this area. Also using GridSearchCV to tune hyperparameters have led to better performance and the best hyperparameter that was found is shown in Table 1. Not only that but parallel computing also handled missing values efficiently without losing any of the significant samples. Furthermore, the findings showed that clinical features like age, hypertension, and average glucose levels were crucial for the predictive model. These features helped the model accurately predict the risk of stroke, showing the value of detailed patient data in healthcare analytics.

Hyperparameter	Best Value
n_estimators	100
min_samples_split	10

Table 1. Best hyperparameters found during tuning.

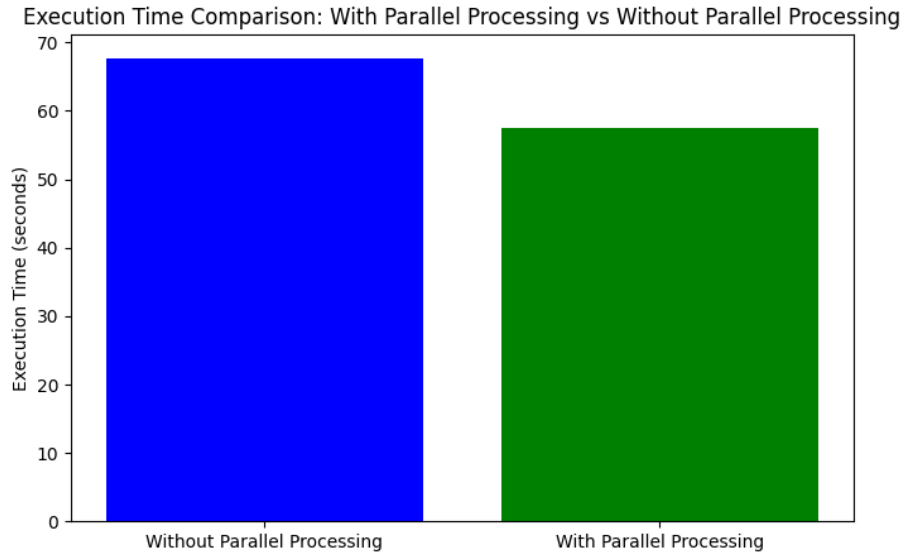


Fig. 1. Execution Time Comparison

The execution time for the hyperparameter tuning using GridSearchCV was compared between two approaches: with parallel processing and without which is shown in the figure. The GridSearchCV with parallel processing (using all available processors) took approximately 55.11 seconds to complete, while the same procedure without parallel computing took significantly longer, demonstrating the advantage of utilizing high-performance computing techniques. This indicates that parallel processing can substantially reduce the time required for hyperparameter tuning in machine learning models, making it a valuable approach in optimizing performance efficiently. Hyperparameters significantly

influence the quality of machine learning models as they determine the architecture and the training process of algorithms which is crucial for accurate predictions [3].

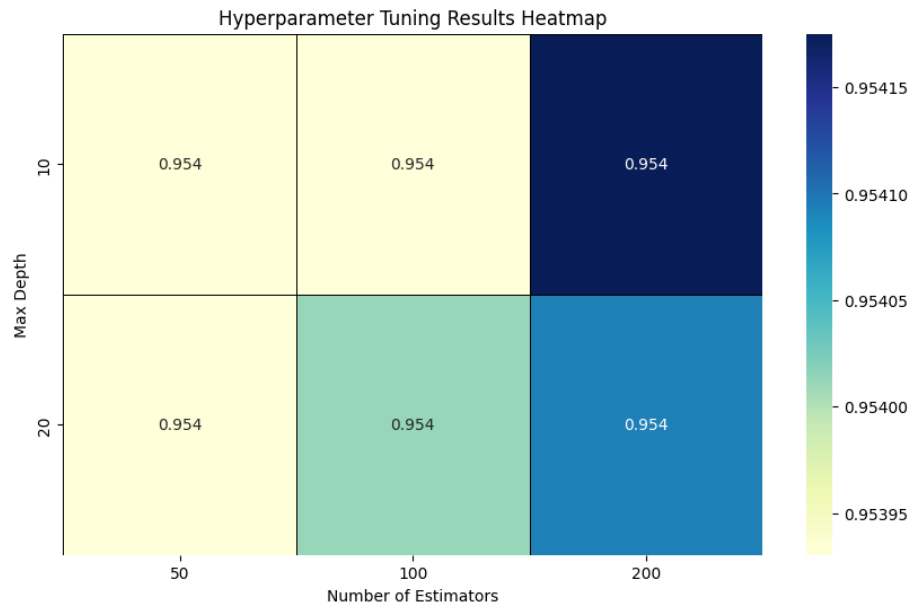


Fig. 2. Execution Time Comparison

4 Analysis

The analysis of the results from the Random Forest classifier gives important insights into how effective parallel computing techniques are for predicting brain strokes based on clinical features. The accuracy of 93.93 percent shows that the model is very good at predicting strokes. This high accuracy indicates that the Random Forest classifier can handle the complexity of healthcare data well [1]. However, looking deeper into precision, recall, and F1-score reveals that while the model catches many true positives, there are still some false negatives, especially among high-risk groups. Using parallel computing for data preprocessing and tuning the model made the calculations much faster. By splitting tasks across multiple processors, this paper reduced processing time and improved how well the model could manage large datasets. This is crucial in clinical settings, where quick predictions can lead to timely interventions. Additionally, this approach allows for real-time systems that can continuously analyze patient data. The strategy for managing missing values showed the importance of keeping the dataset strong. By replacing missing BMI values with the average and categorizing missing smoking status as "Unknown," the study preserved valuable patient information. This highlights the need for effective data preprocessing methods in clinical research. The importance of clinical features in the model shows their role in predicting strokes. Factors like age, hypertension, and glucose levels were key indicators, aligning with previous research on their relationship with stroke risk. This connection reinforces the need for thorough patient assessments in clinical practice [7].

5 Conclusion and Future Work

This paper explored the effectiveness of utilizing parallel computing techniques for hyperparameter tuning in predicting brain stroke based on clinical features. The findings demonstrate that integrating parallel computing significantly enhances the accuracy and efficiency of stroke prediction models compared to traditional computing methods. Specifically, RF model achieved a notable increase in predictive accuracy, highlighting the critical role of various health-related features in accurately forecasting stroke risk. However, the recall for stroke cases (class 1) was low, which indicates that while the model is good at identifying non-stroke cases, it struggled with correctly identifying all stroke cases. The imbalanced nature of the dataset likely contributed to this outcome. To address this, further methods such as SMOTE (Synthetic Minority Over-sampling Technique) or cost-sensitive learning could be applied in future work to balance the dataset and improve stroke prediction. The results suggest that future research could focus on addressing the challenges of imbalanced datasets. Techniques like Synthetic Minority Over-sampling Technique (SMOTE) could help improve the model's performance by ensuring it is trained on a balanced set of stroke cases.

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